Question

Three-dimensional crystals possess translational symmetry in all three directions. However, there exists a class of structures, including quasicrystals, that disrupt this property due to complex reasons. As crystallographers delved deeper, it was discovered that these structures restore translational symmetry when viewed from a higher-dimensional perspective, hence they are referred to as higher-dimensional crystals. Compound $\mathbf{A}$ is a common $(3+1)D$ crystal with the composition $(\mathrm{TTF})\mathrm{I}_{0.7+x}(0<x<0.1)$, where TTF (tetrathiafulvalene) and I exhibit the same periodicity along the a and b directions, while their periodicities differ along the c direction, resulting in a "vernier caliper"-like structure in this direction. If the ratio $c_{\mathrm{T}}/c_{\mathrm{TT}}$ is approximately considered an integer ratio, the combined superstructure possesses a conventional unit cell.

The known superstructure cell parameters of $\mathbf{A}$ are $a=4.815\ \mathrm{nm}$, $b=1.605\ \mathrm{nm}$, $c\approx 2.500\ \mathrm{nm}$, $\beta=91.50°$. If any lattice point of the two substructures is placed at the $(0,0,0)$ position in the superstructure, then TTF lattice points are located at $(1/3,0,0)$, $(5/6,1/2,0)$, $(0,0,1/7)$, while I lattice points are located at $(1/6,0,14/15)$, $(0,1/2,9/10)$, $(0,0,1/5)$ (minor variations in the coordinates of TTF and I can be neglected).

Based on the above information, determine the lattice types of the TTF and I substructures (if only the b-direction is centered, it can be denoted as B-centered, using the superstructure coordinate system as reference).

A. TTF substructure: A-centered monoclinic;
I substructure: A-centered monoclinic

B. TTF substructure: A-centered monoclinic;
I substructure: B-centered monoclinic

C. TTF substructure: A-centered monoclinic;
I substructure: C-centered monoclinic

D. TTF substructure: B-centered monoclinic;
I substructure: A-centered monoclinic

E. TTF substructure: B-centered monoclinic;
I substructure: B-centered monoclinic

F. TTF substructure: B-centered monoclinic;
I substructure: C-centered monoclinic

G. TTF substructure: C-centered monoclinic;
I substructure: A-centered monoclinic

H. TTF substructure: C-centered monoclinic;
I substructure: B-centered monoclinic

I. TTF substructure: C-centered monoclinic;
I substructure: C-centered monoclinic

Answer

Correct Answer: G

Detailed Explanation

Analysis of the TTF Substructure

Known TTF lattice point coordinates (with the supercell as the reference frame):

The translation vectors of the TTF lattice are $(0,0,0),(1/3,0,0),(5/6,1/2,0),(0,0,1/7)$

CHECKPOINT 1 PTS

The translation vectors of the TTF lattice are $(0,0,0),(1/3,0,0),(5/6,1/2,0),(0,0,1/7)$

These points are all lattice points in the TTF lattice. Therefore, the vector difference between them must also be a translation vector of the TTF lattice.

Finding translation vectors in the a, b directions:

The vector from P_0 to P_1 is $v_1 = P_1 - P_0 = (1/3,0,0)$. This indicates that $(1/3,0,0)$ is a translation vector. Then, its integer multiples are also translation vectors, such as $2*v_1 = (2/3,0,0) and 3*v_1 = (1,0,0)$.

The vector from P_0 to P_2 is $v_2 = P_2 - P_0 = (5/6,1/2,0)$.

Now we have two translation vectors v_1 and v_2 in the ab plane.

Calculate $v_2 - v_1 = (5/6,1/2,0) - (1/3,0,0) = (5/6-2/6,1/2,0) = (3/6,1/2,0) = (1/2,1/2,0)$.

CHECKPOINT 1 PTS

There exists a translation vector $(1/2, 1/2, 0)$ in the TTF lattice

We have derived a new translation vector $t = (1/2, 1/2, 0)$. This vector is exactly the centering vector of the C-centered lattice.

CHECKPOINT 1 PTS

C-center corresponds to the translation vector $(1/2, 1/2, 0)$

In addition to translations along the supercell basis vector directions (such as $(1, 0, 0)$, etc.), the TTF substructure's lattice also contains an additional translation symmetry of $(1/2, 1/2, 0)$.

This means that the TTF substructure is C-centered with respect to the supercell. Since the superstructure is monoclinic, the TTF substructure is C-centered monoclinic.

CHECKPOINT 1 PTS

The TTF substructure is C-centered monoclinic

Analysis of the I Substructure

Known translation vectors:

$$u_1 = (1/6, 0, 14/15), u_2 = (0, 1/2, 9/10), u_3 = (0, 0, 1/5)$$

CHECKPOINT 1 PTS

The translation vectors of the I lattice are $(0,0,0),(1/3,0,0),(5/6,1/2,0),(0,0,1/7)$

The basis vectors of the I substructure can be written as a linear combination of the supercell basis vectors. Let the basis vectors of the I substructure be a', b', c', then: $a' = m_{11}a + m_{12}b + m_{13}cb' = m_{21}a + m_{22}b + m_{23}cc' = m_{31}a + m_{32}b + m_{33}c$. The given lattice point coordinates (u, v, w) must be an integer multiple linear combination of these basis vectors. $(1/6,0,14/15),(0,1/2,9/10),(0,0,1/5)$ are all translation vectors. We hope to find whether there exists a translation vector $t = (p,q,r) + n_1a' + n_2b' + n_3c'$, where (p,q,r) is one of $(1/2,1/2,0),(1/2,0,1/2)$, or $(0,1/2,1/2)$. Note that $u_2 = (0,1/2,9/10)$. $u_2 - (0,1/2,1/2) = (0,0,9/10-1/2) = (0,0,4/10) = (0,0,2/5) = 2*(0,0,1/5)$. Therefore, $u_2 - 2*u_3 = (0,1/2,1/2)$. Since u_2 and u_3 are both translation vectors of the I lattice, their integer multiple linear combination $(u_2 - 2*u_3)$ is also a translation vector of the I lattice.

CHECKPOINT 1 PTS

An integer multiple linear combination of translation vectors is also a translation vector

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We have derived the translation vector $t = (0,1/2,1/2)$.

CHECKPOINT 1 PTS

There exists a translation vector $(0,1/2,1/2)$ in the I lattice

This vector is exactly the centering vector of the A-centered lattice.

CHECKPOINT 1 PTS

A-center corresponds to the translation vector $(0, 1/2, 1/2)$

The lattice of the I substructure contains a translation symmetry of $(0, 1/2, 1/2)$.

This means that the I substructure is A-centered with respect to the supercell. Since the superstructure is monoclinic, the I substructure is A-centered monoclinic.

CHECKPOINT 1 PTS

The I substructure is A-centered monoclinic